

A comprehensive review of monkeypox (mpox) on the occasion of the ongoing multicountry outbreak of the disease

Panagiotis Toumasis¹, Vasileios Paparizos², Georgia Vrioni¹

¹Department of Microbiology, Medical School, National and Kapodistrian University of Athens, Athens, Greece.

²Sexually Transmitted Infections Unit, 1st Department of Dermatology and Venereology, Andreas Syggros University Hospital of Athens, Athens, Greece.

269



Summary

The ongoing recent outbreak of mpox, extending to both endemic and non endemic regions, has emerged as a growing international public health threat. The purpose of this review is to highlight the basics of the disease. Mpox is a zoonotic infectious disease, caused by MPV, a DNA virus in the genus *Orthopoxvirus*. There are two possible means of mpox transmission: animal-to-human and human-to-human transmission. In this epidemic, mpox is predominantly affecting men who have sex with men. The early symptoms and signs of mpox are nonspecific but the subsequent evolution of the skin lesions is characteristic. The preferred diagnostic method for mpox is Polymerase Chain Reaction (PCR) testing of samples from skin lesions. The

management of mpox includes supportive care and pain relief medication. There is no treatment approved specifically for mpox. However, there is evidence that antiviral agents that are effective against variola virus are also effective against MPV. There are several simple precautions in order to avoid mpox, including vaccination.



Key words

mpox, outbreak, epidemic

Corresponding author

Georgia Vrioni

Department of Microbiology

Medical School,

National and Kapodistrian University of Athens

75, M. Asias, 115 27, Athens, Greece

Tel. +30 210 7492129

Email: gvrioni@med.uoa.gr

Introduction

The 2022 outbreak of monkeypox (recently renamed as “mpox”¹) extending to both endemic and non-endemic regions is an issue that has alerted the international scientific society, while the World Health Organisation (WHO) declares the spread of the disease a “public health emergency of international concern”. In the shadow of COVID19 pandemic, mpox virus has been considered by some as the next great viral threat.

Mpox virus was first identified in 1958 by Danish virologist Preben von Magnus in monkeys kept for research² but was first identified in humans, specifically in a 9-month-old boy, in 1970 in the Democratic Republic of Congo.³ The virus is endemic in the tropical rainforest regions across Central and West Africa, where sporadic outbreaks have occurred over the past 50 years. Cases that happen outside endemic countries are related to international travel to endemic regions. Since May 2022, multiple cases of mpox have been reported worldwide from non-endemic countries and most of them are associated with trips to countries in Europe and North America, rather than African endemic countries.⁴ Besides, reported mpox cases have involved mainly, but not exclusively, men who have sex with men (MSM).⁵

Mpox case numbers have grown up progressively. So, although cases decreased significantly after September 2022, it is critical for healthcare professionals worldwide to stay up-to-date and be well informed on mpox, since they might encounter this condition in their practice. The purpose of this review is to shed light to the basics of mpox, including its clinical presentation, management and prophylaxis.

Virology

Mpox virus (MPV, MPXV, or hMPXV) is the cause of mpox in both humans and animals. MPV is a zoonotic (transmitted to humans from animals), double-stranded DNA virus, belonging to the *Poxviridae* family, the *Chordopoxvirinae* subfamily, and the genus *Orthopoxvirus*.⁶ Regarding the species that are contained in the *Orthopoxvirus* genus, we should highlight the other 3 species that cause infection in humans: *variola virus* (VARV), the cause of smallpox, *cowpox virus* (CPX), the cause of cowpox and *vaccinia virus* (VACV), which is the source of the modern smallpox vaccine.⁶ Two distinct genetic clades of the mpox virus are described: Clade I, formerly the Congo Basin (Central African) Clade, and Clade II, formerly the West African clade.⁷ The latter consists of two subclades:

Clade IIa and Clade IIb.⁷ In comparison with Clade II, Clade I has been associated with a more severe form of the disease with higher mortality and more evident human-human transmission.⁶ All publicly available MPV genomes from the 2022 mpox outbreak to date belong to Clade II, subclade IIb.⁸

On electron microscopy, MPV appears large as its size ranges from 200 to 250 nanometers.¹⁰ MPV has the same morphological characteristics as other orthopoxviruses: a linear double-stranded DNA genome surrounded by a lipoprotein outer membrane in an oval or brick-shaped appearance.¹⁰ MPV genome is about 197 kb and contains about 190 nonoverlapping ORFs >180 nt long.¹¹

Despite the fact that MPV is a DNA virus –which means that MPV would express its genome in the nucleus of eukaryotic cells– its life cycle takes place in the cytoplasm of infected cells.¹⁰ MPV genome encodes all the proteins that are required for viral DNA replication, including the RNA polymerase system, using the host ribosomes.¹⁰

Transmission path

There are two possible means of mpox transmission: animal-to-human transmission and human-to-human transmission (Figure 1).

Regarding the former, mpox is known to be a zoonosis: a disease that is transmitted from animals to humans.⁹ The animal reservoir for mpox disease is considered to include various species of monkeys and rodents, namely squirrels, rats and dormice, found in African regions.⁹ Zoonotic transmission happens via direct contact with the blood, body fluids, and inoculation from mucocutaneous lesions of an infected animal.¹² The MPV transmission could occur whether the aforementioned natural viral hosts are alive, i.e. via bites/scratches or dead, i.e. handling infected carcasses or consumption of inadequately cooked meat.¹²

Concerning inter-human transmission, it is attributed to respiratory exposure to droplets from infected individuals, which means that prolonged face-to-face contact increases the risk of transmission.¹² In addition, MPV can be transmitted via close contact with skin lesions of infected individuals or recently contaminated objects such as towels or clothing.¹² Also, congenital mpox should be considered since MPV can be transmitted via the placenta from infected mother to fetus or via close contact with the newborn during and after birth.¹³

It should be mentioned that the ongoing epidemic has unique characteristics since the disease is spreading primarily through sexual contact¹⁴ (via direct con-

tact with groin and genital lesions) and is predominantly affecting men who have sex with men.¹⁵ However, it is insufficient to classify mpox as a sexually transmitted disease. In the current epidemic outbreak, the reproduction number R_0 for MPV is estimated to be 2,44, with high variability among European countries and possibly affected by major cluster events of human to human transmission.⁸ This high R_0 indicates that the MPV may persist in the human population. Like other Sexually Transmitted Diseases, e.g. syphilis or warts, MPOX probably not transmitted by semen, but in the vast majority of cases the transmission took place through sexual contact.

Clinical presentation

The time between exposure to mpox virus and appearance of the first clinical features (“incubation period”) has been estimated at 6 to 13 days (range from 5 to 21 days).¹² The early symptoms and signs of mpox are nonspecific, with fever, chills, fatigue, swollen lymph nodes and muscle pains being the most common.¹⁶ Within 1 to 5 days after the first clinical features, skin lesions make their initial appearance.¹⁷ The latter, characteristically, appear on the face before appearing on the trunk then elsewhere across the patient’s body.¹⁷ Very often they appear in the genital area and the anus (Figures 2a, 2b). The evolution of the skin lesions progresses through four stages: macules (lesions with a flat base), papules (slightly raised firm lesions), vesicles (lesions filled with clear fluid), pustules (lesions filled with yellowish fluid), and scabs.¹⁷ (Figure 3). Patients could suffer with pain. Also, they usually have itchy skin in the healing stage of skin lesions.

First on the list of differential diagnosis stands smallpox, since many clinical features of the latter are similar to mpox. Key points for clinicians to distinguish mpox is early lymphadenopathy that usually appears at the same time with the onset of the fever (which is typically absent in smallpox infection) and slower maturation of skin lesions¹⁸ (monkeypox lesions tend to be similar in size and present at the same evolution stage). Except for smallpox, differential diagnosis of mpox includes: chickenpox, generalized vaccinia, disseminated zoster, eczema herpeticum, disseminated herpes simplex, syphilis, yaws, scabies, rickettsialpox, measles, bacterial skin infections and drug-associated eruption.⁹

Mpox is usually a self-limited disease and its duration is estimated at 2 to 4 weeks.¹² The severity of illness can be determined by many factors such as the initial health of the individual i.e. immunocompro-

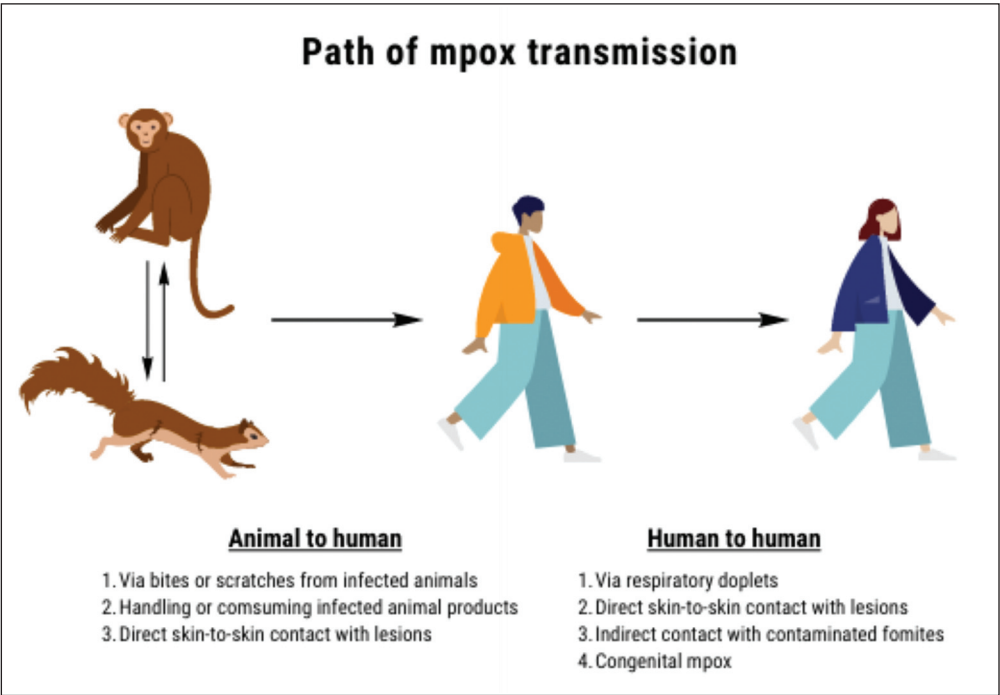


Figure 1 Pathways of mpox transmission.

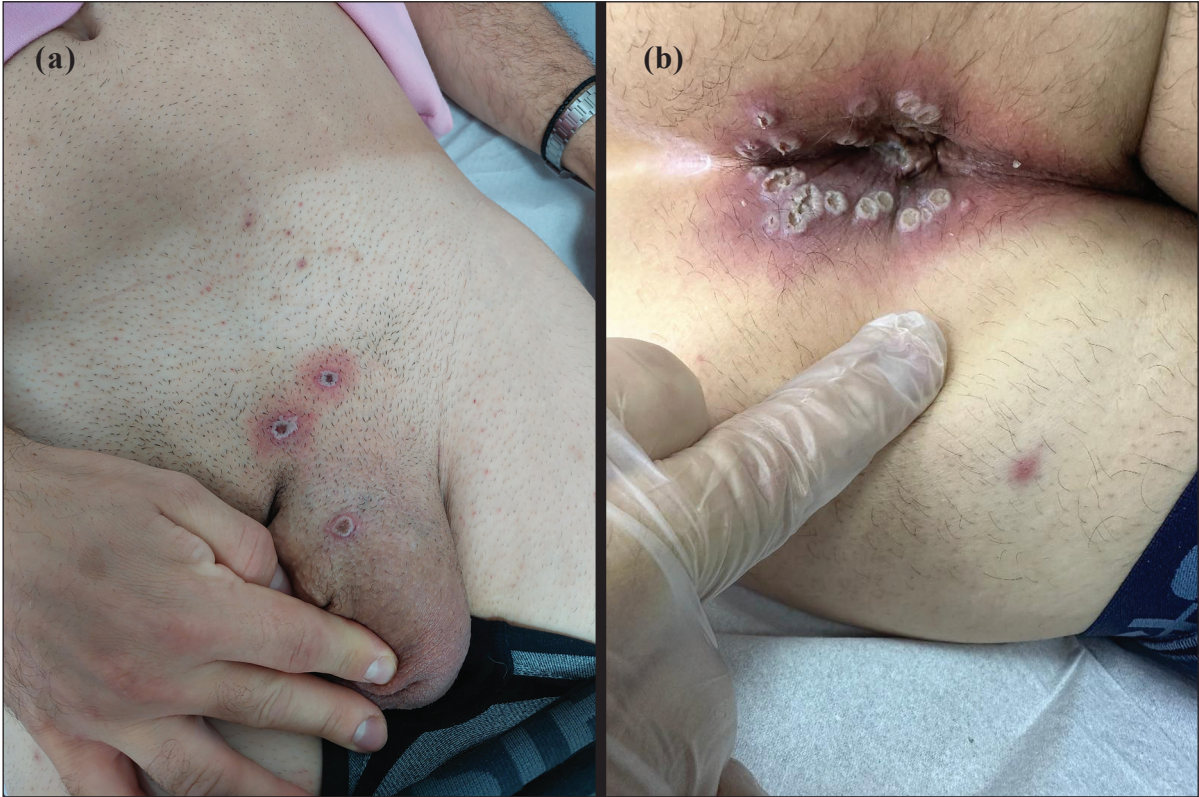


Figure 2 Skin lesions in the genital area (a) and the anus (b).

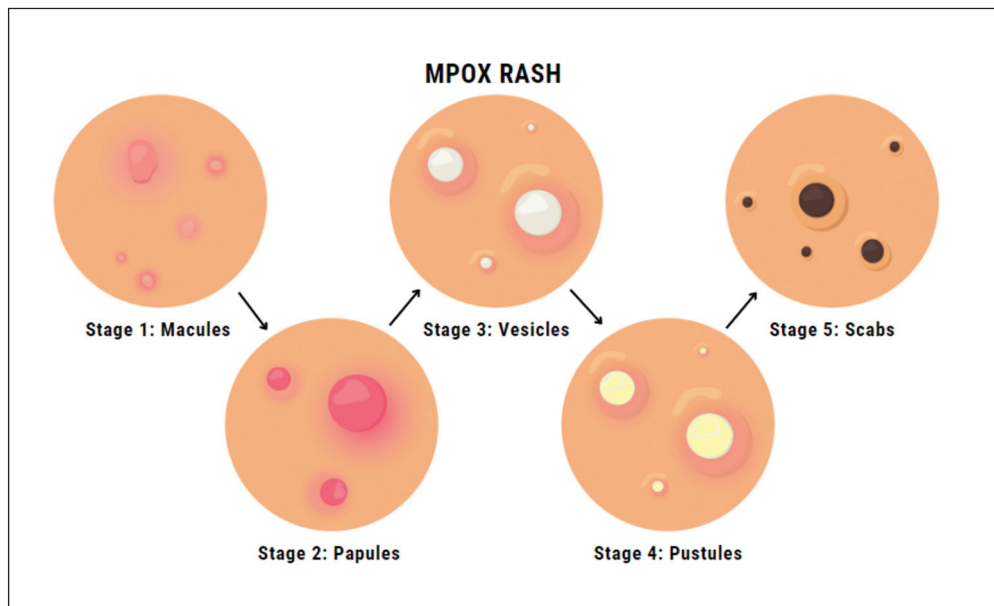


Figure 3 Evolution of the skin lesions due to mpx infection.

mised patients are disproportionately severely affected in comparison with non-immunocompromised patients, but also the clade of mpox virus. Concerning the latter, clade I has been associated with a more severe form of the disease and higher mortality.⁶ Fatality rate for clade II is less than 1% and on the other hand fatality rate for clade I is estimated up to 11% in unvaccinated kids.⁹ Patients suffering with mpox might experience complications such as mouth lesions leading to deteriorated nutrition, secondary bacterial infections of the skin lesions, pneumonitis, conjunctivitis or keratitis and hardly ever sepsis and septic shock. On the contrary, the existence of asymptomatic mpox infection is not impossible.¹⁹ The asymptomatic patients don't know about their status and can transmit the disease to their healthy close contacts.

Diagnosis

Diagnosis of mpox is made on the basis of phenotypic methods, namely medical signs and reported symptoms, and genetic methods. Among diagnostic laboratory tests the golden standard is Polymerase Chain Reaction (PCR), given its high sensitivity and specificity.²⁰ Clinicians should take samples from skin lesions, namely the roof or fluid from vesicles and pustules, and dry crusts, keep them in a dry, sterile tube and send them to a laboratory. PCR blood tests are

usually inconclusive since MPV stays in blood only a short time.

Clinical management

Patients with mild mpox usually recover without medical intervention. However, sometimes they need supportive care or/and pain relief medication.²¹ Regarding the first, supportive care includes maintenance of adequate fluid balance, hemodynamic support, supplemental oxygen, treatment of bacterial superinfections of skin lesions.²¹ Regarding the second, clinicians should consider acetaminophen, NSAIDs and topical steroids and anesthetics such as lidocaine.

Regarding antiviral therapy, there is no treatment approved specifically for mpox. However, there is evidence that antiviral agents that are effective against variola virus are also effective against mpox virus.²¹ Those are *tecovirimat*, *cidofovir* and *brincidofovir*, a prodrug of *cidofovir*, conjugated with a lipid molecule.²¹

Tecovirimat. Clinicians should consider *tecovirimat* as the treatment of choice. *Tecovirimat*, also known as TPOXX or ST-246, is an inhibitor of the orthopoxvirus VP37 envelope wrapping protein, preventing replication and dissemination of the virus into the host.²² In mid-July 2018, the United States Food and Drug Administration (FDA) approved TPOXX as the first drug

with an indication for treatment of smallpox in adults and children,²³ and in January 2022, the European Medicines Agency, also, approved TPOXX for treatment of smallpox and cowpox.²⁴ TPOXX is not approved as a treatment for other orthopoxvirus infections including mpox since there is no data on its efficacy in humans. But, a non-research expanded access Investigational New Drug protocol (EA-IND), held by CDC, permits clinicians to use TPOXX to treat mpox.²⁵ Also, since September 2022, there has been an ongoing clinical trial, named STOMP.²⁶ The latter is a randomized, placebo-controlled, double-blind study to evaluate the effectiveness of tecovirimat for the treatment of people with laboratory-confirmed or presumptive mpox disease.²⁶ Tecovirimat is available as oral capsules (each capsule contains 200mg TPOXX) and injection vials.²⁵ Based on EA-IND protocol, the recommended dose is calculated based on the patient's weight.²⁵ For instance, for patients weighing from 40 kg to < 120 kg, the recommended oral dose is 600 mg (3 capsules) every 12 hours.²⁵ The standard duration of therapy is 14 days.²⁵ The most frequently reported adverse reactions of TPOXX were headache, nausea, abdominal pain and vomiting, if TPOXX was taken orally, and pain, swelling, erythema and extravasation at infusion site and headache, if TPOXX was taken IV.²⁵

Cidofovir / Brincidofovir. In the mid 1996, FDA approved the cidofovir, also known as Vistide, as a treatment option for CMV retinitis in patients with acquired immunodeficiency syndrome (AIDS).²⁷ Cidofovir seems to be effective against orthopoxviruses in vitro and animal studies but there is no data on its effectiveness in human mpox. Brincidofovir, also known as CMX001 or Tembexa, is a prodrug of cidofovir. In mid-June 2021, CMX001 was approved by the FDA for the treatment of human smallpox disease in both adults and children, including neonates.²⁸

Except for antiviral agents, *vaccinia immune globulin (VIGIV)* is another potential treatment of mpox. DFA approved VIGIV for the treatment of complications due to vaccinia vaccination.²⁹ Lately, CDC holds an expanded access IND protocol that allows the use of VIGIV for the treatment of orthopoxviruses (including mpox) in an outbreak.³⁰

Prevention and immunization

According to CDC, individuals should be aware of several simple precautions to reduce their chances of contracting mpox virus.³¹ Those are the following:

1. Avoiding close contact with animals that are able to harbor mpox (rodents and primates) and with individuals who are diagnosed with mpox or have a rash that looks like mpox. Also, avoiding objects that were used by the latter.
2. Having proper hand hygiene.
3. Getting vaccinated

Since the ongoing epidemic is spreading primarily through sexual contact among men who have sex with men (MSM), the latter should adapt some aspects of their sex life, apart the aforementioned protective measures.³² MSM should take a temporary break from sex activities until two weeks after their second dose have passed, limit their number of sex partners, use condoms in order to lower the risk of direct contact with groin and genital lesions etc. There can be no doubt raising public awareness and educational interventions should be of high priority.

There are two available vaccines against mpox: Modified Vaccinia Ankara (MVA) vaccine, known as JYNNEOS in USA, IMVANEX in Europe, and IMVAMUNE in Canada and ACAM2000 vaccine. The first is a third-generation vaccine and contains an attenuated strain of the vaccinia virus. In August 2022, the FDA issued an emergency use authorization for this vaccine to be injected intradermally in individuals aged 18 years and older who are at high risk for mpox infection.³³ The first dose gives some protection against mpox, but in order to provide stronger protection, the vaccine should be administered as two doses given four weeks apart.³⁴ The safety profile of this vaccine seems to be excellent. It can be used individuals with immunocompromising conditions.³³ The second is licensed by FDA to be used against smallpox but it can be used under an expanded-access investigational new drug (IND) application through the CDC.³⁵ The adverse effects associated with ACAM2000 vaccine seems to be more than the adverse effects of MVA vaccine, including myopericarditis/pericarditis. ACAM2000 is contraindicated in immunocompromised individuals.

Conclusion

Nowadays, since we live in a globalized world, we should not ignore neglected tropical diseases. Mpox has been declared as a public health emergency. Individuals all over the world, especially health professionals and high-risk populations, should be concerned and well informed about the current situation, placing emphasis on the preventive measurements, in order to impede the increase of mpox burden.



Περίληψη

Λοίμωξη από ιό της ευλογιάς των πιθήκων (monkeypox, mpox): ανασκόπηση της βιβλιογραφίας με αφορμή τη συνεχιζόμενη, πολυεθνική έξαρση της νόσου

Παναγιώτης Τουμάσης¹, Βασίλειος Παπαρίζος², Γεωργία Βρυώνη^{*1}

¹Εργαστήριο Μικροβιολογίας, Ιατρική Σχολή, Εθνικό και Καποδιστριακό Πανεπιστήμιο Αθηνών.

²Μονάδα Ειδικών Λοιμώξεων, 1η Πανεπιστημιακή Δερματολογική Κλινική, Νοσοκομείο Δερματικών
και Αφροδισίων Νοσημάτων «Α. Συγγρός», Αθήνα.

*Υπεύθυνος αλληλογραφίας

Η έξαρση της λοίμωξης mpox, τόσο σε ενδημικές όσο και σε μη ενδημικές περιοχές, αναδεικνύεται ως μια εξελισσόμενη απειλή για την δημόσια υγεία παγκοσμίως. Σκοπό της παρούσας ανασκόπησης αποτελεί η υπογράμμιση των βασικών σημείων σχετικά με τη λοίμωξη. Η νόσος mpox είναι ζωνόσος, η οποία προκαλείται από τον ιό MPV, έναν DNA ιό που ανήκει στο γένος *Orthoroxvirus*. Υπάρχουν δύο πιθανοί οδοί μετάδοσης: από ζώα στον άνθρωπο και από άνθρωπο σε άνθρωπο. Στη τωρινή επιδημία, η νόσος εξαπλώθηκε κυρίως σε άνδρες που κάνουν σεξ με άνδρες. Τα πρώιμα συμπτώματα και σημεία της νόσου δεν είναι ειδικά αλλά η επακόλουθη ανάπτυξη δερματικών βλαβών είναι χαρακτηριστική. Η διαγνωστική μέθοδος που προτιμάται είναι η αλυσιδωτή αντίδραση πολυμεράσης (PCR) σε δείγματα δερματικών βλαβών. Η διαχείριση της νόσου περιλαμβάνει υποστηρικτική θεραπεία και αναλγητική αγωγή. Δεν υπάρχει ειδική θεραπεία για την νόσο mpox. Εντούτοις, υπάρχουν ενδείξεις πως αντιικές ουσίες που είναι αποτελεσματικές έναντι του ιού της ευλογιάς είναι αποτελεσματικές και για τον ιό MPV. Υπάρχουν αρκετά απλά μέτρα πρόληψης της νόσου mpox, συμπεριλαμβανομένου του εμβολιασμού.



Λέξεις κλειδιά

mpox, έξαρση, επιδημία



References

1. World Health Organization. (n.d.). WHO recommends new name for Monkeypox disease. World Health Organization. Retrieved January 5, 2023, from <https://www.who.int/news/item/28-11-2022-who-recommends-new-name-for-monkeypox-disease>
2. Magnus PV, Andersen EK, Petersen KB, Birch-Andersen A. A pox-like disease in cynomolgus monkeys. *Acta Pathologica Microbiologica Scandinavica* 1959; 46:156-176.
3. Breman JG, Ruti K, Steniowski MV, Zanotto E, Gromyko AI, Arita I, & World Health Organization. (1980). Human monkeypox 1970-1979/by JG Bremen, Kalisa-Ruti, MV Steniowski, E. Zanotto, AI Gromyko and I. Arita.
4. World Health Organization. (n.d.). Mpox (Monkeypox) outbreak 2022. World Health Organization. Retrieved January 5, 2023, from <https://www.who.int/emergencies/situations/monkeypox-oubreak-2022>
5. Iñigo Martínez J, Gil Montalbán E, Jiménez Bueno S, Martín Martínez F, Nieto Juliá A, Sánchez Díaz J, et al. Monkeypox outbreak predominantly affecting men who have sex with men, Madrid, Spain, 26 April to 16 June 2022. *Euro Surveill* 2022; 27:2200471.
6. Lansiaux E, Jain N, Laivacuma S, Reinis A. The virology of human monkeypox virus (hMPXV): A brief overview. *Virus Res* 2022; 322:198932.
7. Gigante CM, Korber B, Seabolt MH, Wilkins K, Davidson W, Rao AK, et al. Multiple lineages of monkeypox virus detected in the United States, 2021-2022. *Science* 2022; 378:560-565..
8. Branda F, Pierini M, Mazzoli S. Monkeypox: Early estimation of basic reproduction number R0 in Europe. *J Med Virol* 2023; 95:e28270.
9. Moore MJ, Rathish B, Zahra F. Mpox (Monkeypox) [Updated 2022 Nov 30]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK574519/>
10. Alakunle E, Moens U, Nchinda G, Okeke MI. Monkeypox Virus in Nigeria: Infection Biology, Epidemiology, and Evolution. *Viruses* 2020; 12:1257.
11. Kugelman JR, Johnston SC, Mulembakani PM, Kisalu N, Lee MS, Koroleva G, et al. Genomic variability of monkeypox virus among humans, Democratic Republic of the Congo. *Emerg Infect Dis* 2014; 20:232-239.
12. World Health Organization. (n.d.). Monkeypox transmission . World Health Organization. Retrieved January 9, 2023, from <https://www.who.int/news-room/fact-sheets/detail/monkeypox>
13. Centers for Disease Control and Prevention. (2022, October 14). Clinical considerations for Mpox in people who are pregnant or breastfeeding. Centers for Disease Control and Prevention. Retrieved January 9, 2023, from <https://www.cdc.gov/poxvirus/monkeypox/clinicians/pregnancy.html>
14. WHO disease outbreak news. Monkeypox - United Kingdom of Great Britain and Northern Ireland. 18 May, 2022. Available at: <https://www.who.int/emergencies/disease-outbreaknews/item/2022-DON383>.
15. Liu X, Zhu Z, He Y, Lim JW, Lane B, Wang H, et al. Monkeypox claims new victims: the outbreak in men who have sex with men. *Infect Dis Poverty* 2022; 11:84.
16. NHS. (n.d.). Monkeypox. NHS choices. Retrieved January 10, 2023, from <https://www.nhs.uk/conditions/monkeypox/>
17. Petersen E, Kantele A, Koopmans M, Asogun D, Yinka-Ogunleye A, et al. Human Monkeypox: Epidemiologic and Clinical Characteristics, Diagnosis, and Prevention. *Infect Dis Clin North Am* 2019; 33:1027-1043.
18. Jezek Z, Szczeniowski M, Paluku KM, Mutombo M, Grab B. Human monkeypox: confusion with chickenpox. *Acta Trop* 1988; 45:297-307.
19. De Baetselier I, Van Dijck C, Kenyon C, Coppens J, Michiels J, de Block T, et al. Retrospective detection of asymptomatic monkeypox virus infections among male sexual health clinic attendees in Belgium. *Nat Med* 2022; 28:2288-2292.
20. Patel M, Adnan M, Aldarhami A, Bazaid AS, Saeedi NH, Alkayyal AA, et al. Current Insights into Diagnosis, Prevention Strategies, Treatment, Therapeutic Targets, and Challenges of Monkeypox (Mpox) Infections in Human Populations. *Life* 2023; 13:249.
21. Centers for Disease Control and Prevention. Treatment Information for Healthcare Professionals. Available at: <https://www.cdc.gov/poxvirus/monkeypox/clinicians/treatment.html>
22. National Center for Biotechnology Information. PubChem Compound Summary for CID 16124688, Tecovirimat. <https://pubchem.ncbi.nlm.nih.gov/compound/Tecovirimat>. Accessed Jan. 2, 2023.
23. U.S Food & Drug Administration . FDA approves the first drug with an indication for treatment of smallpox. 2018. <https://www.fda.gov/news-events/press-announcements/fda-approves-first-drug-indication-treatment-smallpox>
24. European Medicines Agency. Tecovirimat SIGA. 2022. Available at: <https://www.ema.europa.eu/en/medicines/human/EPAR/tecovirimat-siga>.
25. Centers for Disease Control and Prevention. Guidance for Tecovirimat Use Under Expanded Access Investigational New Drug Protocol during 2022 U.S. Monkeypox Cases. Available at: <https://www.cdc.gov/poxvirus/monkeypox/clinicians/Tecovirimat.html>.

26. ClinicalTrials.gov. Study of Tecovirimat for Human Monkeypox Virus (STOMP). <https://clinicaltrials.gov/ct2/show/NCT05534984>
27. FDA approves cidofovir for treatment of CMV retinitis. Food and Drug Administration. *J Int Assoc Physicians AIDS Care*. 1996 Aug;2(8):30. PMID: 11363758.
28. US Food and Drug Administration. FDA approves drug to treat smallpox. Rockville, MD: US Food and Drug Administration; June 4, 2021 (News Release). Available at: <https://www.fda.gov/drugs/news-events-human-drugs/fda-approves-drug-treat-smallpox>.
29. <https://www.fda.gov/media/78174/download>
30. Centers for Disease Control and Prevention. Expanded Access IND Protocol: Use of Vaccinia Immune Globulin Intravenous (VIGIV, CNJ-016) for Treatment of Human Orthopoxvirus Infection in Adults and Children. Available at: <https://www.cdc.gov/poxvirus/monkeypox/pdf/VIGIV-IND-Protocol.pdf>
31. CDC. “Mpox in the U.S.” Centers for Disease Control and Prevention, 31 Oct. 2022, www.cdc.gov/poxvirus/monkeypox/prevention/protect-yourself.html.
32. CDC. “Mpox in the U.S.” Centers for Disease Control and Prevention, 16 Dec. 2022, www.cdc.gov/poxvirus/monkeypox/prevention/sexual-health.html.
33. “Monkeypox Update: FDA Authorizes Emergency Use of JYNNEOS Vaccine to Increase Vaccine Supply.” Monkeypox Update: FDA Authorizes Emergency Use of JYNNEOS Vaccine to Increase Vaccine Supply | FDA, 9 Aug. 2022, www.fda.gov/news-events/press-announcements/monkeypox-update-fda-authorizes-emergency-use-jynneos-vaccine-increase-vaccine-supply.
34. Kuehn BM. Single Monkeypox Vaccine Dose Provides Some Protection. *JAMA* 2022; 328:1801.
35. CDC. “ACAM 2000 Vaccine.” Centers for Disease Control and Prevention, 21 Oct. 2022, www.cdc.gov/poxvirus/monkeypox/interim-considerations/acam2000-vaccine.html.